

Math 343 / 643 Spring 2026

Midterm Examination Two

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Full Name _____

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signature

date

Instructions

This exam is 75 minutes (variable time per question) and closed-book. You are allowed **one** page (front and back) of a “cheat sheet”, blank scrap paper (provided by the proctor) and a graphing calculator (which is not your smartphone). Please read the questions carefully. Within each problem, I recommend considering the questions that are easy first and then circling back to evaluate the harder ones. No food is allowed, only drinks.

Problem 1 Consider the following full-rank design matrix:

$$\mathbf{X} := [\mathbf{1}_n \mid \mathbf{x}_{\cdot 1} \mid \dots \mid \mathbf{x}_{\cdot p}] = \begin{bmatrix} \mathbf{x}_{1\cdot} \\ \vdots \\ \mathbf{x}_{n\cdot} \end{bmatrix}$$

with column indices $0, 1, \dots, p$ and row indices $1, 2, \dots, n$. And let $\mathbf{H}$ be the orthogonal projection matrix (which is idempotent and symmetric) onto the column space of $\mathbf{X}$. We assume also a continuous (real-valued) response model which is linear in these measurements, i.e. $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\mathcal{E}}$. For the error term, we assume the “core assumption”,

$$\boldsymbol{\mathcal{E}} \sim \mathcal{N}_n(\mathbf{0}_n, \sigma^2 \mathbf{I}_n) \quad \text{where } \sigma^2 > 0 \text{ and thus } \frac{1}{\sigma^2} \boldsymbol{\mathcal{E}}^\top \boldsymbol{\mathcal{E}} \sim \chi_n^2.$$

Consider the following estimator for $\boldsymbol{\beta}$: $\mathbf{B} := (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$ and let $\hat{\mathbf{Y}} := \mathbf{X}\mathbf{B}$ and $\mathbf{E} := \mathbf{Y} - \hat{\mathbf{Y}}$. We established in class that $\mathbf{B} \sim \mathcal{N}_{p+1}(\boldsymbol{\beta}, \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1})$. We also established that $\mathbf{B}$ and $\mathbf{E}$ are independent. Let $\boldsymbol{\epsilon}$ denote the realization from $\boldsymbol{\mathcal{E}}$.

- (a) [2 pt / 2 pts] Is $\mathbf{B}$ an unbiased estimator for $\boldsymbol{\beta}$? Circle one: **Yes** / No
- (b) [2 pt / 4 pts] Is $\mathbf{B}$ the maximum likelihood estimator for $\boldsymbol{\beta}$? Circle one: **Yes** / No
- (c) [2 pt / 6 pts] Is $\mathbf{E}$ an unbiased estimator estimator for σ^2 ? Circle one: Yes / **No**
- (d) [2 pt / 8 pts] Does $\mathbb{E}[\boldsymbol{\mathcal{E}} - \mathbf{E}] = \mathbf{0}_n$? Circle one: **Yes** / No
- (e) [5 pt / 13 pts] Write the z test statistic for testing $H_0 : \beta_{17} = 37$ assuming σ^2 is known.

$$\begin{aligned} & \mathbf{B} \sim \mathcal{N}_{p+1}(\boldsymbol{\beta}, \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}) \\ \Rightarrow & B_{17} \sim \mathcal{N}(\beta_{17}, \sigma^2 (\mathbf{X}^\top \mathbf{X})_{17,17}^{-1}) \\ \Rightarrow & \frac{B_{17} - \beta_{17}}{\sigma \sqrt{(\mathbf{X}^\top \mathbf{X})_{17,17}^{-1}}} \sim \mathcal{N}(0, 1) \\ \Rightarrow & z = \frac{b_{17} - 37}{\sigma \sqrt{(\mathbf{X}^\top \mathbf{X})_{17,17}^{-1}}} \end{aligned}$$

- (f) [5 pt / 18 pts] Find the distribution of $\hat{\mathbf{Y}}$ from the information given in the problem header and results from MATH 340. Simplify as much as possible.

$$\hat{\mathbf{Y}} := \mathbf{X}\mathbf{B} \sim \mathcal{N}_{\dim[\mathbf{X}\mathbf{B}]} \left((\mathbf{X})\boldsymbol{\beta}, \sigma^2(\mathbf{X}) (\mathbf{X}^T \mathbf{X})^{-1} (\mathbf{X}^T) \right) = \mathcal{N}_n (\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{H})$$

- (g) [8 pt / 26 pts] Prove that $\boldsymbol{\mathcal{E}}^\top (\mathbf{I}_n - \mathbf{H}) \boldsymbol{\mathcal{E}} = \text{SSE}$. Justify each step.

$$\begin{aligned} \boldsymbol{\mathcal{E}}^\top (\mathbf{I}_n - \mathbf{H}) \boldsymbol{\mathcal{E}} &= \boldsymbol{\mathcal{E}}^\top (\mathbf{I}_n - \mathbf{H}) (\mathbf{I}_n - \mathbf{H}) \boldsymbol{\mathcal{E}} \quad (\text{since } \mathbf{I}_n - \mathbf{H} \text{ is idempotent}) \\ &= \boldsymbol{\mathcal{E}}^\top (\mathbf{I}_n - \mathbf{H})^\top (\mathbf{I}_n - \mathbf{H}) \boldsymbol{\mathcal{E}} \quad (\text{since } \mathbf{I}_n - \mathbf{H} \text{ is symmetric}) \\ &= ((\mathbf{I}_n - \mathbf{H}) \boldsymbol{\mathcal{E}})^\top (\mathbf{I}_n - \mathbf{H}) \boldsymbol{\mathcal{E}} \quad (\text{linear algebra identity}) \\ &= \|(\mathbf{I}_n - \mathbf{H}) \boldsymbol{\mathcal{E}}\|^2 \quad (\text{linear algebra identity}) \\ &= \|(\mathbf{I}_n - \mathbf{H})(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})\|^2 \quad (\text{substitution}) \\ &= \|(\mathbf{I}_n - \mathbf{H})\mathbf{Y} - (\mathbf{I}_n - \mathbf{H})\mathbf{X}\boldsymbol{\beta}\|^2 \quad (\text{algebra}) \\ &= \|\mathbf{E} - \mathbf{0}_n\|^2 \quad (\text{symmetries from MATH 342W}) \\ &= \mathbf{E}^\top \mathbf{E} = \sum_{i=1}^n E_i^2 \quad (\text{linear algebra identity}) \\ &= \text{SSE} \quad (\text{by definition}) \end{aligned}$$

Problem 2 Consider the Boston Housing Data which has $n = 506$ and response `medv` with $\bar{y} = 22.53$ and $s_y = 9.20$. We assume the linear model is true and model `medv` using OLS on `zn + rm + nox + dis + lstat`, all continuous (non-categorical) features. Below is the $(\mathbf{X}^T \mathbf{X})^{-1}$ where $\mathbf{X}$ is the design matrix:

	(Intercept)	zn	rm	nox	dis	lstat
(Intercept)	0.58000	4.4e-04	-5.1e-02	-2.7e-01	-1.8e-02	-3.0e-03
zn	0.00044	6.9e-06	-4.5e-05	-7.1e-05	-5.1e-05	-2.0e-07
rm	-0.05100	-4.5e-05	6.9e-03	7.5e-04	6.3e-04	4.4e-04
nox	-0.27000	-7.1e-05	7.5e-04	4.2e-01	1.5e-02	-1.9e-03
dis	-0.01800	-5.1e-05	6.3e-04	1.5e-02	1.5e-03	4.3e-05
lstat	-0.00300	-2.0e-07	4.4e-04	-1.9e-03	4.3e-05	8.9e-05

The RMSE for this regression is 5.289 and here are the slope estimates:

(Intercept)	zn	rm	nox	dis	lstat
16.14	0.06	4.44	-15.20	-1.44	-0.66

- (a) [5 pt / 31 pts] Assume that errors are independent. Given the information above, would you be able to create $\hat{CI}_{\beta_{\text{nox}}, 95\%}$, the confidence interval for the true slope parameter of the variable `nox`? If yes, then compute it below and indicate if the CI is approximate or exact. If no, explain why.

No. In this assumption regime, we would need to use the bootstrap. The bootstrap involves resampling with replacement and fitting OLS each time. That information is not given above.

- (b) [5 pt / 36 pts] Assume that errors are independent, mean zero and the errors are homoskedastic. Given the information above, would you be able to create $\hat{CI}_{\beta_{\text{nox}}, 95\%}$, the confidence interval for the true slope parameter of the variable `nox`? If yes, then compute it below and indicate if the CI is approximate or exact. If no, explain why.

$$\hat{CI}_{\beta_{\text{nox}}, 95\%} \approx \left[b_{\text{nox}} \pm z_{97.5\%} s_e \sqrt{(\mathbf{X}^T \mathbf{X})_{\text{nox}, \text{nox}}^{-1}} \right] = \left[-15.20 \pm 1.96 \cdot 5.289 \sqrt{0.42} \right] = [-21.92, -8.48]$$

- (c) [5 pt / 41 pts] Assume that errors are independent, mean zero and the errors are heteroskedastic. Given the information above, would you be able to create $\hat{CI}_{\beta_{\text{nox}}, 95\%}$, the confidence interval for the true slope parameter of the variable **nox**? If yes, then compute it below and indicate if the CI is approximate or exact. If no, explain why.

No. Under the assumption of heteroskedasticity, we would need to use the Hubert-White robust sandwich estimator. This would involve using the residuals $e_1, \dots, e_n$ to compute a diagonal matrix. We were not given that information above.

- (d) [5 pt / 46 pts] Assume that errors are *normal*, independent, mean zero and the errors are heteroskedastic. Given the information above, would you be able to create $\hat{CI}_{\beta_{\text{nox}}, 95\%}$, the confidence interval for the true slope parameter of the variable **nox**? If yes, then compute it below and indicate if the CI is approximate or exact. If no, explain why.

No for the same reason in the previous question.

- (e) [5 pt / 51 pts] Assume the core assumption. But now consider that the estimates in the problem header came from a ridge regression with $\lambda = 1$. Given the information above and the facts from problem 1, would you be able to create $\hat{CI}_{\beta_{\text{nox}}, 95\%}$, the confidence interval for the true slope parameter of the variable `nox`? If yes, then compute it below and indicate if the CI is approximate or exact. If no, explain why.

No. Because we are not given hardly any of the information required to find the distribution of the `nox` slope coefficient estimator which is one dimension of the following vector rv:

$$\begin{aligned} \mathbf{B}_{\text{ridge}} &= (\mathbf{X}^T \mathbf{X} + \mathbf{I}_n)^{-1} \mathbf{X}^T \mathbf{Y} = (\mathbf{X}^T \mathbf{X} + \mathbf{I}_n)^{-1} \mathbf{X}^T (\mathbf{X} \boldsymbol{\beta} + \boldsymbol{\varepsilon}) \\ &= (\mathbf{X}^T \mathbf{X} + \mathbf{I}_n)^{-1} \mathbf{X}^T \mathbf{X} \boldsymbol{\beta} + (\mathbf{X}^T \mathbf{X} + \mathbf{I}_n)^{-1} \boldsymbol{\varepsilon} \\ &\sim \mathcal{N}_{p+1} \left((\mathbf{X}^T \mathbf{X} + \mathbf{I}_n)^{-1} \mathbf{X}^T \mathbf{X} \boldsymbol{\beta}, \sigma^2 (\mathbf{X}^T \mathbf{X} + \mathbf{I}_n)^{-1} (\mathbf{X}^T \mathbf{X} + \mathbf{I}_n)^{-1} \right) \end{aligned}$$

- (f) [5 pt / 56 pts] Assume the $\mathbf{X}$ matrix's columns were standardized so that the average of each column was zero and the standard deviation of each column was 1. Let $\lambda = 1000$. Would the slope coefficient estimates found in the problem header be expected for a lasso regression fit? Why or why not?

No. After standardization, a large $\lambda = 1000$ will likely force the variables to zero or near zero in a lasso.

- (g) [5 pt / 61 pts] Assume the core assumption. Compute $\mathbb{P}(R^2 > 0.9)$ under H_0 that all features in the model `zn`, `r`, `nox`, `dis`, `lstat` do not have any linear effect. You will need to express your answer using `pbeta`(x, α, β), the CDF of the beta distribution. But each argument (i.e. x, α, β) must be numeric.

We learned in class that $R^2 \mid H_0 \sim \text{Beta}\left(\frac{p}{2}, \frac{n-(p+1)}{2}\right)$ thus substituting in the direction of the integrated area, the value of $n = 506$ and the value of $p = 5$,

$$\mathbb{P}(R^2 > 0.9 \mid H_0) = 1 - \text{pbeta}(0.9, 2.5, 250)$$

Problem 3 Consider the following survival distribution:

$$Y \sim \text{Lomax}(\alpha, \beta) := \frac{\alpha}{\beta} \left(1 + \frac{y}{\beta}\right)^{-(\alpha+1)} \mathbb{1}_{y>0}$$

We will assume a restricted parameter space $\alpha > 2$, $\beta > 0$ so that the expectation and variance always exists:

$$\mathbb{E}[Y] = \frac{\beta}{\alpha - 1}, \quad \text{Var}[Y] = \frac{\beta^2 \alpha}{(\alpha - 1)^2 (\alpha - 2)}$$

We would like to reparameterize the Lomax so that $\mathbb{E}[Y] = \theta > 0$. We do so below:

$$Y \sim \text{Lomax}(\alpha, \theta) := \frac{\alpha}{\theta(\alpha - 1)} \left(1 + \frac{y}{\theta(\alpha - 1)}\right)^{-(\alpha+1)} \mathbb{1}_{y>0}$$

We now wish to use a GLM to estimate p slope coefficients β using data from a matrix $\mathbf{X}$ defined in the problem 1 header.

- (a) [5 pt / 66 pts] What is an example link function ϕ we can use to connect θ to $\mathbf{x}_i \cdot \beta$ and why would your example work in this context?

$$\theta = \phi(\mathbf{x}_i \cdot \beta) = e^{\mathbf{x}_i \cdot \beta}$$

This works because $\mathbf{x}_i \cdot \beta \in \mathbb{R}$ and thus $e^{\mathbf{x}_i \cdot \beta} > 0$ which is the parameter space of θ .

Regardless of your answer to the previous problem, employ $\theta = \phi(\mathbf{x}_i, \boldsymbol{\beta}) = e^{\mathbf{x}_i \cdot \boldsymbol{\beta}}$ going forward. Also assume $Y_1, \dots, Y_n \stackrel{iid}{\sim} \text{Lomax}(\alpha, \theta)$ as defined in the problem header.

- (b) [7 pt / 73 pts] Write an expression below that would compute the $\mathbf{B}$, maximum likelihood estimator (not estimate) for vector $\boldsymbol{\beta}$ and A , the maximum likelihood estimator (not estimate) for α as a function of $\mathbf{x}_1, \dots, \mathbf{x}_n, \mathbf{Y}$. You do not need to solve. You do not need to simplify. Assume there is *no censoring*. There are many equivalent expressions that answer this question correctly.

$$\{\mathbf{B}, A\} := \arg \max_{\boldsymbol{\beta} \in \mathbb{R}^p, \alpha > 2} \left\{ \prod_{i=1}^n \frac{\alpha}{e^{\mathbf{x}_i \cdot \boldsymbol{\beta}} (\alpha - 1)} \left(1 + \frac{Y_i}{e^{\mathbf{x}_i \cdot \boldsymbol{\beta}} (\alpha - 1)} \right)^{-(\alpha+1)} \right\}$$

- (c) [3 pt / 76 pts] Are there any nuisance parameter(s) in this estimation? If yes, list the nuisance parameter(s). If no, write “none”.

α

We employ the correct MLE for $\mathbf{B}, A$ going forward.

- (d) [2 pt / 78 pts] If there was censoring in $Y_1, \dots, Y_n$ would the MLE expression in (b) change? Circle one: **Yes** / No
- (e) [2 pt / 80 pts] Is A asymptotically normal? Circle one: **Yes** / No
- (f) [2 pt / 82 pts] Is $\text{Var}[A] = \text{Var}[Y]$? Circle one: Yes / **No**

We now fit the lung survival data on the uncensored observations. The response variable is days of survival. We find the following results:

	Value	Std. Error	z	p
(Intercept)	7.17e+00	1.08e+00	6.64	3.2e-11
inst	2.05e-02	8.80e-03	2.32	0.0201
age	-7.54e-03	8.06e-03	-0.94	0.3497
sex	3.91e-01	1.39e-01	2.82	0.0048
ph.ecog	-6.26e-01	1.58e-01	-3.95	7.7e-05
ph.karno	-1.86e-02	7.67e-03	-2.43	0.0151
pat.karno	7.68e-03	5.54e-03	1.39	0.1656
meal.cal	7.32e-06	1.81e-04	0.04	0.9678
wt.loss	1.10e-02	5.34e-03	2.07	0.0387
Loglik(model)= -824.2 Loglik(intercept only)= -841.1				

- (g) [6 pt / 88 pts] Given the information above, would you be able to create $\hat{CI}_{\beta_{\text{age}}, 95\%}$, the confidence interval for the true slope parameter of the variable **age**? If yes, then compute it below to three significant digits and indicate if the CI is approximate or exact. If no, explain why.

$$\hat{CI}_{\beta_{\text{age}}, 95\%} \approx [b_{\text{age}} \pm z_{97.5\%} s_{b_{\text{age}}}] = [-0.00754 \pm 1.96 \cdot 0.00806] = [-0.0233, 0.00826]$$

- (h) [7 pt / 95 pts] Test the omnibus null at the 5% significance level. Justify each step. Here are some values of the inverse CDF of the χ^2_{df} distribution:

df	Probability less than the critical value				
	0.90	0.95	0.975	0.99	0.999
1	2.706	3.841	5.024	6.635	10.828
2	4.605	5.991	7.378	9.210	13.816
3	6.251	7.815	9.348	11.345	16.266
4	7.779	9.488	11.143	13.277	18.467
5	9.236	11.070	12.833	15.086	20.515
6	10.645	12.592	14.449	16.812	22.458
7	12.017	14.067	16.013	18.475	24.322
8	13.362	15.507	17.535	20.090	26.124

Since $\hat{\Lambda} = 2(\ln(\mathcal{L}_{\text{full}}) - \ln(\mathcal{L}_{\text{reduced}}))$ and these log likelihood values are given in the parameter estimate table, we then calculate $\hat{\Lambda} = 2(-824.2 - -841.1) = 33.8 > \chi^2_{8,95\%} = 15.507$ as determined from the χ^2 table above since we have 8 features we are zeroing out, we reject H_0 .

- (i) [5 pt / 100 pts] Predict the number of days of survival for someone with $\text{inst} = 1$, $\text{age} = 75$ but all other measurements are zero.

$$\hat{y}_\star = e^{\mathbf{x}_\star \mathbf{b}} = e^{b_0 + b_{\text{inst}}(1) + b_{\text{age}}(75)} = e^{7.17 + 0.0205(1) + -0.00754(75)} = e^{6.625} = 753.7 \text{ days}$$